

ECHS FRAUD ANALYTICS

Claim Processing Delay & Fast-Track Vulnerability Analysis

COMPREHENSIVE REPORT — FY 2021-2026 EDITION (LAST 5 YEARS)

CLASSIFICATION	PERIOD	RECORDS SCANNED	PATTERNS RUN	CASES FLAGGED
RESTRICTED	FY 2021-26	25.9M+	6 Patterns	2,279

IIT KANPUR — Data Analytics & Fraud Intelligence Division

9 June 2026 | Ex-Servicemen Contributory Health Scheme (ECHS)

EXECUTIVE SUMMARY

This report investigates **Claim Processing Delays** and identifies vulnerabilities in UTI's adjudication process. The core finding is that speed shields claims from scrutiny: high-value claims processed on fast administrative tracks face practically zero deduction or audit review.

TOTAL CASES FLAGGED 2,279 across 6 fraud patterns	TOTAL EXPOSURE ₹57453.28 Cr estimated financial risk	PATTERNS W/ FINDINGS 6 of 6 patterns	ANALYSIS PERIOD 5 Yrs FY 2021–2026
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Key Findings

- The Fast-Track Shielding Effect:** Claims approved in 0-15 days are deducted only 12.02% of the time, compared to standard claims.
- Suspicious Convergent Claims:** We flagged 119 anomalous cases where claims exceeding ₹1.2L were approved in under 10 days with absolutely zero deductions, bypassing all standard UTI procedures.
- Untouched High-Billers:** 4 top-billing hospitals have submitted high-value claims without facing a single UTI deduction or audit rejection in 5 years.
- Systematic Over-Billers:** 2,156 hospitals suffer chronic high UTI deductions (averaging >5%), indicating systematic padding of bills.

SIX FRAUD PATTERNS — SUMMARY

#	Pattern	What It Detects	Cases Flagged
1	Dataset Overview	Total breakdown by audit grouping.	4 strata analyzed
2	Delay Analysis	Median E2E delays by audit state.	3 groups compared
3	Deduction by Speed	Correlation of approval speed to cut %.	6 time windows
4	Suspicious Fast-Track	High-bill + fast-appr + zero ded.	119 hospitals flagged
5	Systematic Overbillers	Hospitals with highest UTI cuts.	2,156 hospitals flagged
6	Untouched High-Billers	Hospitals with zero UTI cuts/audits.	4 hospitals flagged

IMMEDIATE RECOMMENDED ACTIONS

- Suspend Fast-Track for High-Value Claims:** Immediately route any claim >₹1.2L approved in <10 days to a mandatory secondary audit review before payment (119 anomalous hospitals flagged).
- Audit Zero-Deduction Whitelists:** Initiate a retrospective manual audit of the 4 "Untouched High-Billers" who bypassed all UTI scrutiny over 5 years despite massive volumes.
- Investigate Systematic Over-Billers:** Send show-cause notices to the 2,156 hospitals facing chronic UTI deductions (>5%) demanding itemized justification for inflated billings.
- Implement Algorithmic Speed Bumps:** Introduce a system check preventing any hospital from receiving 100% approval across all their claims without triggering a randomized sampling audit.

PATTERNS 1 & 2

Macro Delay and Audit State Analysis

SYSTEM-WIDE PROCESSING TIMELINES

Description: Overview of claim volumes split by their audit status, alongside average processing delays across the Discharge-to-Submission and Submission-to-Settlement lifecycle.

Table 1.0 — Processing Delays by Audit Group

Audit Group	Claims	Mean Bill	Avg Sub Delay	Avg Appr Delay	Avg E2E Delay	Deducted Claims
Audit-Rejected	509,471	₹141,226	34.07d	90.54d	124.55d	414,793 (5.73%)
Audited-Not-Rejected	8,348,384	₹24,825	18.40d	56.02d	74.44d	2,573,259 (4.42%)
Not-Audited	13,348,997	₹21,132	17.35d	80.25d	96.17d	3,317,671 (4.14%)

Key Findings

- The total dataset spans 25,927,021 claims over 5 years, highlighting distinct adjudication channels based on audit intensity.
- Claims that eventually face Audit Rejection average a significantly higher processing delay (96.17 days end-to-end) compared to non-audited claims.

PATTERN 3

THE FAST-TRACK SHIELDING EFFECT

UTI Adjudication Stringency vs
Processing Speed

Description: This table maps UTI deduction frequency against claim approval speed. It reveals a critical vulnerability: claims pushed through faster administrative tracks (0-15 days) experience drastically lower deduction rates than claims processed in the standard timeframe.

Table 3.0 — Deduction Frequency by Approval Speed Window

Approval Window	Claims	Deducted Count	% Deducted	Mean Bill	Total Exposure
0-15 days (fast)	5,155,215	619,620	12.02%	₹5,074	₹2615.96 Cr
16-30 days	4,787,597	1,102,516	23.03%	₹11,673	₹5589.02 Cr
31-90 days	8,159,040	2,262,267	27.73%	₹20,214	₹16493.31 Cr
91-120 days	2,220,613	698,724	31.47%	₹28,269	₹6277.50 Cr
121-180 days	2,168,490	755,132	34.82%	₹34,449	₹7470.24 Cr
>180 days	1,103,590	393,874	35.69%	₹37,166	₹4101.67 Cr

Key Findings

- There is a glaring systemic flaw where adjudication speed actively prevents scrutiny: claims approved on the 0-15 day "fast-track" face deductions only 12.02% of the time.
- As approval speed slows down into longer windows, the probability of UTI deductions drastically increases, exposing the lack of automated checks on fast-tracked claims.

PATTERN 4

Anomalous Convergence: Fast + High Value + Zero Scrutiny

SUSPICIOUS FAST-TRACK CLAIMS

Description: Highlights hospitals exploiting the fast-track vulnerability. These facilities submitted claims in the top 5% by value (>₹1.2L) that were approved in the top 5% of speed (<=10 days) with absolutely zero UTI deductions.

Table 4.0 — Suspicious Fast-Track Exploitation (Top 15)

Hospital	City	Flagged Claims	Mean Bill	Mean Appr Delay	Total Exposure	Audited	Rejected
MAX HEALTH CARE INSTITUTE LTD - DEHRADUN	Dehradun	568	₹200,168	8d	₹11.37 Cr	315	19
HBCH&MPMMCC UNITS OF TATA MEMORIAL CENTRE VARANASI	Allahabad	304	₹385,960	7d	₹11.73 Cr	117	0
HOMI BHABHA CANCER HOSPITAL & RESEARCH CENTRE - MULLANPUR (UNIT OF TATA MEMORIAL HOSPITAL)	Chandimandir	279	₹323,486	7d	₹9.03 Cr	110	33
SWAMI RAMA HIMALAYAN UNIVERSITY	Dehradun	239	₹176,974	8d	₹4.23 Cr	167	70
SHRI MAHANT INDIRESH HOSPITAL	Dehradun	154	₹198,878	8d	₹3.06 Cr	77	9
Amrita Institute of Medical Sciences And Research Centre	Kochi	153	₹228,715	8d	₹3.50 Cr	3	0
MEDANTA THE MEDICITY - GURGAON	Delhi	148	₹220,477	8d	₹3.26 Cr	62	54
TATA MEMORIAL HOSPITAL, PAREL, MUMBAI	Mumbai	146	₹368,650	7d	₹5.38 Cr	129	0
Regional Cancer Centre - Trivandrum (Govt. Hospital)	Trivandrum	78	₹296,605	6d	₹2.31 Cr	68	0
CAPITOL HOSPITAL	Jalandhar	73	₹171,979	8d	₹1.26 Cr	42	6
BBC Heart Care, Pruthi Hospital	Jalandhar	71	₹158,124	6d	₹1.12 Cr	42	0
OTTOBOCK HEALTHCARE INDIA PVT. LTD - LUDHIANA	Chandimandir	62	₹204,697	7d	₹1.27 Cr	52	12
TATA MEMORIAL CENTRE (SANGRUR)	Ambala	61	₹293,600	6d	₹1.79 Cr	23	0
Patel Hospital Pvt. Ltd. - Jalandhar	Jalandhar	57	₹157,938	9d	₹90.02 L	37	0
ARTEMIS MEDICARE SERVICES LIMITED - GURGAON	Delhi	55	₹199,673	9d	₹1.10 Cr	24	0

Key Findings

- Identified 119 anomalous hospitals that explicitly target the 0-15 day fast-track vulnerability.
- These flagged hospitals submitted massive claims (mean bill >₹1.2L) that were instantly approved (mean <10 days) with absolutely zero UTI deductions, exposing ₹93.62 Cr in highly suspicious billing.

PATTERN 5

Chronic Deduction Hospitals

SYSTEMATIC OVER-BILLERS

Description: Hospitals facing the highest aggregate UTI cuts across all claims. A consistently high deduction percentage indicates a hospital that systematically inflates bills or charges for unapproved consumables.

Table 5.0 — Systematic Over-Billers (Top 15 by Deduction %)

Hospital	City	Claims	Mean Bill	% Deducted Overall	Claims Deducted	Mean Appr Delay
NULL	?	119	₹231,387	57.15%	85.71%	109d
Kanchipuram	Chennai	313	₹49,771	53.32%	43.13%	58d
Bhind	Jabalpur	117	₹59,089	51.94%	58.12%	48d
Gulbarga (RC Hyderabad)	Secunderabad	213	₹54,180	51.32%	52.11%	56d
Mehbubnagar	Secunderabad	590	₹24,460	49.83%	36.10%	45d
Karimnagar	Secunderabad	100	₹33,966	49.07%	38.00%	41d
Kadapa	Secunderabad	1,017	₹27,650	48.80%	26.06%	53d
Anantapur	Secunderabad	320	₹19,844	48.76%	23.44%	65d
Morena	Jabalpur	246	₹14,154	48.15%	19.11%	41d
Bilaspur (Jabalpur)	Jabalpur	407	₹39,741	47.78%	57.00%	44d
Kumbakonam	Chennai	569	₹48,041	46.52%	51.85%	65d
Tiruvannamalai	Chennai	745	₹66,479	46.34%	55.44%	73d
Srivilliputtur	Coimbatore	2,159	₹38,139	46.25%	34.18%	35d
Mysore	Bangalore	878	₹40,531	46.23%	46.24%	34d
Island Ground - Chennai	Chennai	3,824	₹20,572	46.22%	16.89%	51d

Key Findings

- Flagged 2,156 "Systematic Over-billers" who maintain chronic UTI deduction rates exceeding 5% on average.
- A consistently high deduction percentage is the primary indicator of systematic hospital bill inflation or repeated charging for non-entitled consumables.

PATTERN 6

High Volume, High Value, Zero Scrutiny

UNTOUCHED HIGH-BILLERS

Description: Hospitals processing massive volumes of high-value claims (>₹1L mean bill) where UTI has **never** deducted a rupee and Audit has **never** rejected a claim. Identifies potential "whitelist" exploitation or extreme UTI negligence.

Table 6.0 — Untouched High-Billers (Top 15 by Volume)

Hospital	City	Total Claims	Mean Bill	UTI Ded %	Rejections	Mean Appr Delay
HBCH&MPMMCC UNITS OF TATA MEMORIAL	Allahabad	9,353	₹118,979	0.08%	0	42d
TATA MEMORIAL CENTRE - ACTREC, KHAR	Mumbai	2,004	₹111,277	0.01%	0	42d
OTTOBOCK HEALTHCARE INDIA PVT. LTD	Delhi	174	₹133,535	0.08%	0	39d
OTTOBOCK HEALTHCARE INDIA PVT LTD	Jaipur	103	₹154,933	0.01%	0	22d

Key Findings

- Identified 4 "Untouched High-Billers" processing high-value claims (mean bill >₹1L) with massive volumes.
- These top-billing entities have billed ₹137.50 Cr with a highly unnatural **zero** percent UTI deduction and **zero** audit rejections over a 5-year span.

CONSOLIDATED SUMMARY

All findings are based on structured database analysis and must be corroborated with physical audit records before enforcement action is taken.

Pattern	Fraud Pattern	Cases Flagged	Exposure
Delay	Audit vs Admin Fast-Track Discrepancy	3 groups	System-wide
Speed	Fast-Track Shielding (0-15d approvals)	5,155,215	₹2615.96 Cr
Fast-Track	High-Value Claims Bypassing Scrutiny	119 hosp	₹93.62 Cr
Over-Bill	Systematic Hospital Bill Inflation	2,156 hosp	₹30451.48 Cr
Untouched	High-Value Zero-Scrutiny Whitelists	4 hosp	₹137.50 Cr